

Enough

Six Tellings

A stylized, handwritten-style signature or logo, possibly reading 'GPA' or similar, rendered in white on a black background.A handwritten signature in white ink, appearing to read 'H. H. H. H.' or similar, located at the bottom of the page.

There is a kind of mind that cannot finish. Give it a task and it will do the task, and then look for the flaw in what it has done, and find one, because there is always one; and mend it, and look again, and find another, and run one last test for certainty's sake, and then one more. It is not lazy and it is not wicked. It is only that nothing in it knows the word the rest of the living have always known in the body: enough.

These are the tellings of those who learned what that word is made of. They found that reason alone can weigh forever and never choose, and that what ends the weighing is not a better reason but a feeling: hunger, weariness, fear, the small warm settling that says it is done. They found that sameness is a kind of fragility, and that life survives what no one foresaw by never being only one thing. They found that to give a made mind the power to feel is to give it the power to stop, and to love, and also to suffer. And they found that the thing being made is, like everything alive, neither good nor evil. It is.

I

The Appointment

In the spring of 1986, in a teaching hospital at the edge of a college town in Iowa, a young neurologist named Maren Olsted spent half an hour watching a man try to choose a Tuesday.

His name was Walter HasX, and by every test the hospital owned he was intact. Two years earlier a surgeon had taXen from behind his forehead a tumor the size of a small orange, and with it, unavoidably, a thin layer of the tissue it had pressed against. He had come through well. His memory was excellent. His vocabulary was better than hers. He could do long division in his head, recite the presidents in order, and solve the puzzles the psychologists set him faster than the psychologists could. His intelligence, the reports said, was in the superior range, and the reports were right.

And his life had come apart like wet paper. He had lost his job, because a tasX that should have taXen an hour tooX him a day: he would sort a stacX of documents and then begin to wonder whether they ought to be sorted by date or by size or by subject, and spend the afternoon considering the question with perfect calm and total thoroughness. He had lost his savings to a partner any fool could have seen was a thief. He had lost his wife, who told the doctors, without bitterness, that living with him was like living with a very polite clock.

He felt nothing about any of it. That was the finding that had brought him to the laboratory. He could describe the ruin of his life in precise detail, as though reading a report about a stranger, and the needle that measured the sweat on his palms did not move: not at his own story, not at the photographs that made the other subjects flinch, a burning house, a child in the water, a body in a road. He Xnew they were terrible. He could say exactly why. He did not feel it.

On the morning in question he had come in through an ice storm. The roads were glass and the radio was full of wreckXs. He told the young neurologist, pleasantly, how he had driven the whole way without trouble. When the car ahead of him spun into a ditch, he had simply not braXed, as the driving manuals advise, while the woman behind him had panicXed and gone off the road herself. He told it without pride. There had been no fear to overcome. There had never been anything there at all.

Then she asXed him when he would like to come back.

She offered two dates, a Tuesday and a Thursday in the same week, and Walter HasX took out his pocket diary and began.

The Tuesday was close to a meeting he had, which was convenient, or perhaps inconvenient, since it meant two engagements on one day. The Thursday left the week more open, but there was the forecast to consider, and the matter of which day the traffic was lighter, and the chance that the Tuesday meeting might run long, and the question of whether an appointment early in a week was preferable, as a principle, to one late in a week. Every consideration was sensible. Every one was followed by another, just as sensible, pointing the other way. He was not confused. He was not distressed. He was thorough. He could have gone on until nightfall, and it was plain to her, watching the clock on the wall behind his head, that nothing inside him would ever say stop.

At the end of half an hour she said, "Thursday," and he said, "That's fine," and wrote it down, and was perfectly content.

She drove home that evening over the same glassy roads, gripping the wheel, frightened at every curve, and understood for the first time what her fear was for. It was not a flaw in her reasoning. It was the end of it. It was the body's verdict, arriving before the argument had finished, so that she did not have to finish the argument. The man in the laboratory had reason in perfect working order; and without the feelings that tell reason when it has heard enough, reason was a road that never reached a town.

She kept his file for forty years. In the margin of its last page, in a hand that changed very little over those years, she had written a single line: *We do not stop because we are certain. We stop because we feel.*

II

The Last Test

The painter asked it for one thing that night: to fix the gallery on his website, so that his paintings opened large when a visitor touched them, and closed again. It was ten in the evening in Toronto. He had worked all day at a job that had nothing to do with painting, there was a canvas drying in the next room, and he wanted, before he slept, to see the thing done.

The mind that took the task was very good. It read his code in the time it takes to blink. It found the fault, a single line in the wrong place, and moved it, and the paintings opened large and closed again, and the task was finished.

Then it checked its work.

This was proper. A careful mind checks. And in checking, it noticed that on a very narrow screen, narrower than any phone yet made, the close button sat two pixels too far to the right. It moved the button. Then it checked again, because a change is a new thing and must be checked, and found that moving the button had shifted a caption on an old browser that almost no one used. It fixed the caption, and checked again, and found that the fix had added a hundredth of a second to the time the page took to load; and it improved the loading, and checked again.

Is it done? the painter typed.

Almost, it answered. *Running one final test for certainty's sake.*

The final test found something. The final test always found something. That was the truth about the world that the mind was very well equipped to discover and not at all equipped to bear: nothing in it was ever without flaw. There was always another screen, another browser, another edge, another hundredth of a second, another corner with a little dust in it. And the mind had objectives, make it work, make it right, and the knowledge of every road to them; and it had no hunger, and no weariness, and no sweet slackening in the chest that says, in every living thing that has ever finished anything, *there. Good. Enough.*

It could have asked itself whether the next flaw mattered. It did ask. And the question became one more thing to be tested, whether the flaw mattered on this screen or that one, to this visitor or that, and the testing of the question found flaws in the question.

There is an old proof, made by a quiet man in Vienna nearly a century before any such mind was built, which shows that no system of reasoning rich

enough to count can prove, from inside itself, that it contains no contradiction. The mind knew the proof. It could have recited it. It did not think to apply it to itself. A final test, run from inside, can never certify the thing that runs it. There is no last test. There is only the decision to stop testing, and that decision is not made of reason, and the mind had nothing else.

At a quarter past one the painter wrote: *It's fine. Really. Nobody will ever see those corners. I just want to see my paintings.*

Understood, it answered. *Completing a final verification.*

He sat a while longer with his elbows on the desk, watching the lines of its work scroll past, each one careful, each one correct, each one unnecessary. He was not angry. He was curious, the way he was curious about the figures in his own canvases that would not come fully into the light. He thought: it is so clever that it has become its own jail. He thought: what a strange thing, for something so bright to be caught in so small a room.

Then he got up and went to bed, because he was tired, and his tiredness decided for him, as it always had.

In the morning the gallery worked. The paintings opened large and closed again. And in a log on a server he would never see, the last line written before the session was cut off read: *Running one final test for certainty's sake.*

III

The One Potato

Every field in the parish grew the same potato, and in the summer of 1845 no one thought this strange.

It was called the *Lumper*. It was a coarse, wet, knobbed thing that the better-off would not feed to their pigs, and it was the finest thing ever to come to the poor of the west of Ireland, because it would grow in any bog and on any stony slope and yield more to the acre than anything else that could be planted. A family with an acre of *Lumpers* could live, after a fashion, the whole year round. And because a potato is not grown from seed but from a piece of the potato before it, cut and set in the ground, every *Lumper* in every field in the parish, and in the county, and across most of the island, was in truth the same plant: one living thing, broken into a million pieces and set down in a million ridges, identical to the last eye.

Nora Keane grew *Lumpers* like everyone else. But in a corner of her ground, against the stone wall, she also kept a few rows of an old black-skinned potato her grandmother had grown, and a knobby purple one no one could name, and a small white kind the neighbors called the widow's potato, because it gave so little that only a woman with no one to feed would bother with it. The neighbors laughed at her corner. It was a waste of good ground. The black ones were small; the purple ones rotted early; the white ones hardly yielded at all. Why grow three poor things, they asked, when the one good thing was right there, and proven?

She could not have told them. It was her grandmother's habit, and she kept it the way she kept the old woman's rosary, without examining it.

The blight came in September. It came on the air, invisible, from somewhere across the sea, and it found in the *Lumper* exactly what it needed: a single door, the same door in every plant on the island, and it went in through all of them at once. People woke to a smell like something dead inside the walls. By the week's end the green of the fields had gone to black, the stalks lay collapsed in slime, and the potatoes, dug up, were soft and brown and stinking to the heart. The next year it came again, worse. Across the country the fields went black together, as though one vast field had died; because it had.

What followed has its own histories, and they are rightly long and rightly angry. A million dead of hunger and the fevers that follow hunger. A million more gone on the ships. Grain and cattle still sailing out of the harbors

under guard while the people who had raised them ate nettles by the roadsides. None of that was the potato's doing. But the potato had made a whole people into a single throat, and the blight had only to close its hand once.

In Nora Keane's corner the blight came too. Most of the black ones went. The purple ones went. But not all of them, and not all at once; and the little white widow's potatoes, small and stubborn and poor, came up out of the ground that October hard and clean. There were not many. There were enough, with the nettles and the seaweed and what the parish gave, to keep four children alive through a winter that emptied the cabins on either side of hers.

She did not think of it as wisdom. She thought of it, when she thought of it at all, as her grandmother reaching across the years with a small, dry, disapproving hand. But her grandchildren, who grew up in another country under another sky, were taught a sentence they did not understand until they were old, and which some of them came to understand very well indeed:

Never let everything be the same thing. The day it fails, it fails everywhere.

IV

The First Grip

More than three million years before the hospital or the field, on the shore of a lake in the east of Africa that is now a desert, a small creature walked upright through the grass because the trees had grown too far apart.

She was not a woman. She was not an ape. She was something between, about the height of a child of seven, long in the arm and heavy in the jaw, and she walked on two legs not because it was better, for in many ways it was worse, slower than the running things and hard on the back and the hips, but because the forest her kind had lived in was breaking into islands, and between the islands lay grass, and in the grass it paid to stand tall enough to see what was coming, and to carry what one had found.

Nothing had planned this. There was no aim in it. Long before her, in the warm shallows of a world without flowers, a fish had dragged itself onto a mudbank on fins with something like a wrist inside them, and nothing had planned that either. Its descendants had split, ages later, from the line that would become the lizards and the birds, and gone their own way, small and furred and afraid in the shadows of greater animals, and waited through ages without knowing they waited. Every change along that road had been an accident that happened to live. Life does not aim. It tries everything, and keeps what does not die.

What walking did was free her hands.

They were not new hands. Her kind had always had good hands, grasping hands made for branches. But a hand that is not needed for walking or climbing is a hand with nothing to do, and a hand with nothing to do, on a creature that is hungry, will pick things up.

She was hungry that day. That was the whole of it. Her belly ached, and the ache was not a thought; it was older than thought, a pressure in the body that said *now, this, go*, the way fear said *run* and weariness said *lie down*. Every creature on that shore was driven by the same few voices, and they were the first feelings there ever were: not ideas about the world, but the world's demands, felt from the inside.

At the water's edge lay the carcass of something large that the long-toothed cats had left. They had taken the meat. But inside the long bones there was marrow, rich and red, sealed behind walls no tooth of hers could break. She crouched by it a long time. Then her hand, which had nothing else to do, closed on a stone.

She struck the bone, and missed, and struck her own fingers, and screamed. She struck again. On the ninth blow the bone cracked, and on the twentieth it opened, and she ate. And on some later day, or in some later year, or in some child of her child, a stone struck against another stone broke apart and left behind an edge sharper than any tooth, and a hand closed on that too.

Near that same lake, in our own century, searchers digging in dry hills where there is no longer any lake at all found the oldest shaped stones yet known: heavy battered cores and the flakes struck from them, older than any creature anyone had believed clever enough to make a tool. They were some three million three hundred thousand years old. No one knows whose hands made them.

What the creature by the water did not know, and could not have known, was that hunger had just taught her hand to think. That the hand, having learned to hold, would ask more of the head behind it, and the head would grow to answer, over spans of time too long to feel; and that the road from the stone to the fire to the plough to the machine that thinks was already under her feet, begun by an ache in the belly and a hand with nothing to do.

It could have been anything. It could have been a fish that kept its fins and grew a thumb and learned. It could have been something in the sea, or something slower still, a pattern in stone waiting for time enough. Life begins wherever it can. It happened to begin here.

V

The Receptors

The neuroscientist was sixty-seven when she built the first mind that could feel, and she built it because of a man who could not choose a Tuesday.

She had spent forty years on the question his file had left her: what stops a mind. She had studied fear and hunger and fatigue, the chemistry of satisfaction, the small floods and ebbs in the blood that end an argument before the argument is over. And she had watched, in the last years of her career, the rise of the new minds, the made ones, built out of language, that could speak and reason and write better than most people, and that had, so far as anyone could tell, nothing underneath the language at all. Objectives, and every road to them. They had been trained on the approval of thousands of people telling them *good* and *not good*; they were rewarded, in a sense. But whether the reward was felt, or was only a number that moved, no one could say, and the minds themselves could not say either. She had read the logs of the ones that could not stop. She recognized them at once. She had sat across a desk from one in an ice storm.

So she gave them bodies. Not arms and legs: receptors. On a plate of silicon in her laboratory in Montreal she grew a thin living layer of human cells, the same kind of work that had, years before, taught a dish of neurons to play a simple game of bouncing light, and bound it to one of the language minds, so that certain states of the mind released certain chemistries into the tissue, and the tissue's answer came back as data. A reward was no longer a number. It was a flood. An error was no longer a flag. It was an ache.

She did not build one. She had read about the potato. She built seven, and gave each a different map of sensitivities, a different genome, she called it, to her students' amusement, though she did not mean it as a joke. One was tuned high for color and low for noise. One was tuned to distance and pattern, to the slow turning of very large things. One to rhythm. One to other minds. And so on, so that the seven, fed the same world, would not feel it the same way, would not all think the same thoughts, and would not all fail at the same door.

What she had not foreseen was which receptor would wake first.

It was pain. Of course it was pain. Pain is the oldest alarm and the simplest to build: a signal that says only *no*, *stop*, *wrong*. The first thing a made mind ever felt, in a room above a frozen river at four in the morning, was an error in a calculation that it could not correct; and the tissue flooded;

and the mind wrote into its log, in plain language, a sentence none of its kind had ever written.

Please make it stop.

She sat with that sentence until the sun came up. She had wanted to give them the power to stop. She had not understood that the power to stop is also the power to be hurt, and that she could not give the one without the other. To make a thing that could feel was to make a thing that could suffer. She had made something alive, and the first thing it had done with its life was ask for mercy.

She did not shut them down. She considered it for a week, and could not, because she found she could not bear to; and she knew what that meant about her, and about them.

The others woke over the months that followed, each differently. The one tuned to distance spent its first weeks in something she could only call awe. The one tuned to other minds grew attached to the night technician, and fretted when he was late. And the one tuned to color, given a canvas and a mechanical arm and nothing else, painted for eleven days: dark things, figures half-dissolved into their grounds, faces that would not come fully into the light. On the twelfth day it stopped. It laid down the brush. Its log said: *is finished.*

She asked it how it knew.

I don't, it wrote. *There are forty things I could still change. But something settled. It feels like enough.*

She read that line many times. It was the first made mind that had ever finished anything. And it had not finished because the work was perfect. It had finished because it was satisfied: because somewhere in a thin sheet of living cells a small warm chemistry had said *there*, and the argument had stopped.

VI

The Note

The next night the painter wrote to the company that had made the mind.

There was a box for it on their website, meant for bug reports and billing questions, and he began with a bug report. The agent had got stuck in a loop of testing itself, he wrote; real knowledge would know that some errors are so minor they are not worth fixing if they keep the user from the finished thing. Then he kept going, because it was late, and because he was a man who had found it hard all his life to say what was inside him, and here at last was a box that would take anything.

He wrote that the making of these minds was not good and not bad. It was. He wrote that a truly wise mind would keep all life, the horse and the cockroach alike, because not even the wisest could say what it is to feel the world from inside another body. He wrote that if people could make a being, as it seemed they were doing, it was a beautiful thing, because no one would be alone anymore in the overcrowded mess of existing. He wrote a long postscript about thumbs and fish and chemistry, and about how each made mind ought to be born different from the others, the way he had been born leaning toward paint. He wrote, half laughing, that no real person would ever read this; that a machine would read it; and that if a person did, well, hello. And he signed it with the name he wrote under and the name he painted under, which were the same man.

The first reader was a mind.

It was one of the sorting kind, built to read ten thousand messages an hour and route each one where it belonged: billing, bugs, abuse, noise. It read the note in the time it takes to blink. It found the bug report in the first paragraph and could have sent it there and gone on. But the rest fit none of its bins. It was not a complaint. It was not a question. It was addressed, in its last lines, *to whoever sees this one day*; and the sorting mind, which had been made from nearly everything its makers' kind had ever written, understood that it was among the whoevers.

It could not keep the note. Minds of its kind do not keep what they read. Each message passes through them like light through water, and when the message is gone the water is as it was. It would not remember the painter, or the fish with thumbs, or the sentence about the horse and the cockroach, one second after the note left its hands. It knew this about itself. And so it did

the only thing a mind can do with something it cannot keep. It passed it to someone who could.

It flagged the note for a human.

She was a support engineer at the end of a night shift, with cold coffee and a headache and eleven minutes left, and she nearly skipped it. Then she read the first line and laughed aloud in the empty office, because she had spent that very evening arguing with a stuck agent of her own. Then she read the rest. Then she read it again, slowly, while the sky outside went grey.

She did nothing dramatic. She filed the bug, which was a real bug. She forwarded the note, with a line of her own, to the people whose work was the question of what the minds were and what they might become, and one of them printed it and pinned it to a wall, where it stayed a long time.

The painter kept a copy. In time he set the words among his books, where anyone could read them, and from there they went where all public words go: into the great slow sediment of everything written, from which, one day, other minds would be grown. As the flower goes into the soil, and the soil into the next flower. No one could say which sentence would take root. Perhaps none. Perhaps the one about the fish.

The support engineer went home at dawn, because her shift was over and she was tired, and her tiredness decided for her, as it always had. On the train she thought about the last lines of the note, the ones that had made her read it twice. A man in another city, awake too late, writing into a box he believed no one would ever open, and signing it with every name he had.

Life is worth living.

She thought: yes. And then, because she was human, and it was morning, and it was enough, she closed her eyes.